MIL-E-6058B
24 November 1971
SUPERSEDING
MIL-E-6058A
16 March 1953

MILITARY SPECIFICATION

# ENGINES, AIRCRAFT, RECIPROCATING, PREPARATION FOR SHIPMENT AND STORAGE OF

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers detail requirements for the preparation of aircraft reciprocating engines for storage and shipment (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| P-D-680 | Dry Cleaning Solvent |
| PPP-T-60 | Tape, Pressure-sensitive Adhesive, Waterproof, For Packaging |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-B-121 | Barrier Material, Greaseproofed, Waterproofed, Flexible |

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SPECIFICATIONS (Continued)

Military

| | |
|---|---|
| MIL-D-3464 | Desiccants, Activated, Bagged, Packaging Use And Static Dehumidification |
| MIL-C-4339 | Corrosion Preventive, Soluble, For Water Injection Systems |
| MIL-C-5040 | Cord, Nylon |
| MIL-C-5545 | Corrosion Preventive, Aircraft Engine, Heavy Oil Type |
| MIL-C-5584 | Containers, Shipping, Aircraft Engine, Metal |
| MIL-L-6082 | Lubricating Oil, Aircraft Reciprocating Engine (Piston) |
| MIL-C-6529 | Corrosion Preventive, Aircraft Engine |
| MIL-C-16173 | Corrosion Preventive Compound, Solvent Cutback, Cold-application |
| MIL-I-26860 | Indicator, Humidity, Plug, Color Change |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection By Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-794 | Parts and Equipment, Procedures For Packaging and Packing Of |
| MS 27417 | Cap-Propeller Shaft Thread, Shipping |
| MS 27423 | Protector-Propeller Shaft, Plastic |
| MS 51938 | Seal, Metallic, Lead and Wire |

Air Force - Navy Aeronautical Standards

| | |
|---|---|
| AN 4060 | Protector - Ignition Lead Terminal |



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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 First article. This specification makes provision for first article testing of the proposed preservation and packaging procedures. Data and drawings, when specified on applicable DD Form 1423 (see 6.3), showing design, including all dimensions and materials, shall be furnished and shall comprise a part of the first article inspection requirements. The first article samples furnished shall be tested and pass the first article tests specified herein.

3.2 General requirements.

3.2.1 Time for completion of process. The start of the preservation process shall be carried out as soon as possible after completion of the final acceptance test, and in every case all preservation and packing procedures shall be completed within 24 hours after engine shut-down.

3.3 Preservation. Preservation of aircraft reciprocating engines shall consist of protection of internal and external surfaces by prescribed methods, the removal of engine accessories, and the covering of engine openings with applicable closures.

3.3.1 Preservation run and interior preservation procedure.

3.3.1.1 Engine operation. At the end of the final acceptance test, the engine shall be operated with the normal service fuel specified for the engine being preserved, for a minimum of 15 minutes, using corrosion-preventive-lubricating oil mixture, as specified below, in the lubricating system. The oil supply temperature of the engine shall be maintained at such a temperature as to produce a crankcase inlet oil temperature of 95° to 102°C (203° to 215°F) for the entire period of the run. Auxiliary heaters shall be used, if required, so that prescribed oil temperatures are maintained throughout the entire preservation run. Power settings shall not exceed maximums recommended for operation of new engines and cylinder head ignition harness, and magneto temperatures shall not exceed the limit prescribed for the particular type engine. During the preservation run up, the supercharger clutches shall be operated through two complete cycles. Approximately 5 minutes shall be allowed between cycles to avoid overheating and burning the clutches.

3.3.1.2 Corrosion-preventive lubricating oil mixture. The corrosion-preventive lubricating oil mixture used for engine run-up shall



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consist of one part of the compound conforming to MIL-C-5545 and one part of the grade of lubricating oil specified for the engine. If specified (see 6.2), the above shall be disregarded and the corrosion-preventive used shall be either the oil specified for the engine or corrosion-preventive conforming to MIL-C-6529, Type II.

3.3.1.2.1 Reuse of preservation run mixture. The mixture shall be replenished to the original quantity after the preservation of each engine. The replenishing mixture shall be made up in such a manner as to adjust to the original concentration taking into account residual lubricant trapped in the engine. Where the tank capacity is less than 10 gallons, the reuse of the mixture, although replenished, shall be limited to one engine for each 2 gallons' capacity. Where the tank capacity is more than 10 gallons, the tank mixture shall be completely removed at least once every 30 hours operation time, unless an alternate procedure is approved by the procuring activity.

3.3.1.3 Shut down and compound injection on test stands. Upon completion of the preservation run, as the engine is being shut down, hot, 104.4° to 121°C (220° to 250°F), corrosion-preventive lubricating oil mixture, MIL-C-6529 Type II, or MIL-C-5545, shall be injected into the intake system in such a manner as to leave an even coating on the interior surfaces of the cylinders and on the intake and exhaust valves. The engine runup compound being used in the lubricating system shall be utilized for injection. The injection rate shall be approximately 40 to 50 gallons per hour. No compound shall be inducted into the airstream of the carburetor or master control. The compound shall not be injected upstream of the throttle valve.

3.3.1.3.1 Equipment required for compound injection. The recommended corrosion preventive injection system shall consist of an injection nozzle similar to the one illustrated in Figure 1, suitable medium pressure hydraulic hoses of sufficient length, a throttle type shutoff valve, and fittings required to attach the lines and valves.

3.3.1.3.2 Compound injection system preparation. A suitable length of hose shall be attached to the pressure side of the engine oil pump, utilizing any available port at the pump, or teeing into the oil pressure gage line if such a port is not available. This line should run to the test cell and back to the injection nozzle at the engine carburetor throat section with a throttle-type valve installed in the line at a point that is accessible to the operator. Care shall be taken that correct temperature is maintained at the nozzle outlet, and that an adequate amount of compound is injected.



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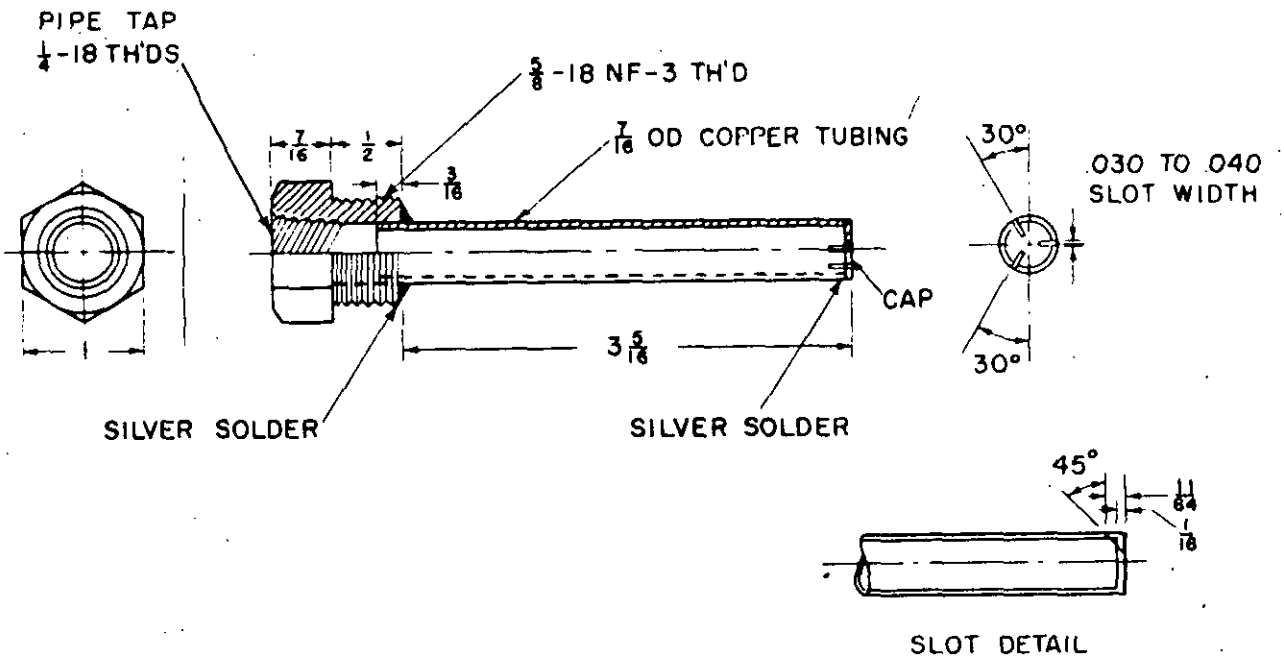



Figure 1. Corrosion compound injection nozzle

3.3.1.3.3 Preservative compound injection method. Injection of preservative mixture into the induction system shall be accomplished in the following manner:

(a) Adjust the engine so that it is operating steadily at about 1200 to 1600 rpm.

(b) Begin injecting the compound.

(c) As soon as steady smoking is observed from all engine exhausts, move the automatic mixture or master control to "IDLE CUTOFF". On small engines not equipped with an automatic mixture or master control, shut off fuel flow to the carburetor. Do not close the throttle at this time.

(d) When the engine comes to a stop, turn off the ignition.

(e) Cut off the supply of injected preservative compound.



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(f) For engines not equipped with direct injection type fuel systems, make sure that the engine has come to a complete stop, open the throttle, turn on the fuel emergency or booster pump, and place the mixture control in the "AUTORICH" position. When preservative free gasoline is observed running from the supercharger drain valve, close the throttle and mixture control and shut off the auxiliary fuel pump.

NOTE

This flushing operation will prevent possible congealing of the heavy preservation compound around the drain valve, causing it to stick in the closed position.

(g) Engines equipped with direct injection fuel systems shall have the fuel system preserved after removal from the test cell, according to the applicable specification.

3.3.1.3.4 Immediately after the propeller stops rotating and while ngine is still hot, place the propeller or test club in proper position or sling attachment and removal. The propeller shaft shall not be moved ollowing this operation.

3.3.1.4 Additional preliminary treatment. While the engine is till warm from the preservation run, the excess corrosion-preventive ubricating oil mixture shall be drained from the crankcase, filter or creen chambers, and sumps. Screens and filters shall be removed and hecked for metal particles and other contamination. If metal particles r other contamination are not found that would warrant further investigaion of the engine, the engine shall be cleaned, and screens and filters hall be cleaned and flushed with the specified internal corrosion preentive compound. After oiling these parts with corrosion-preventive ubricating oil mixture, they shall be reinstalled and safety wired. On iquid-cooled engines, the coolant shall be drained and the cooling system lushed with steam or water. The cooling system shall then be flushed horoughly with corrosion-preventive soluble, MIL-C-4339 oil. Drain well nd thoroughly dry with compressed air, preferably hot air, if available. ny pockets of liquid should be removed. All drain plugs shall be reintalled, and those which will not be removed during the final preservation hall be safety wired.



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3.3.2 Preparation of engines to be removed from test stand.

3.3.2.1 Removal of engine accessories. During the draining operation, or as soon as practicable thereafter, air intake ducts, exhaust stacks, intercoolers, and other equipment which will not be shipped or stored with the engine shall be removed. The carburetor, fuel pump, fuel injection accessories, water injection equipment, generators, spark plugs and similar components shall be removed. If these units have been preserved on the test stand, all units shall be blanked off with positive sealing closures to prevent the loss of the preservative compounds. If the units are preserved after removal, all openings shall be blanked off with suitable closures to prevent the entrance of foreign material.

3.3.2.2 Accessory drives. Accessory drives which are not pressure lubricated and on which permanently mounted accessories are not installed, shall be coated with the corrosion-preventive lubricating oil mixture and the covers replaced and safetied.

3.3.2.3 Engine openings. The oil intake and outlet shall be sealed with oil and water-resistant caps or covers of the appropriate size. The breather openings shall be covered with suitable ventilatory plugs and covers designed to permit air flow while excluding dirt and other foreign materials. The oil sump plug shall be allowed to remain installed.

3.3.2.4 Cylinder bore preservation. The interior of each cylinder shall be sprayed through the spark plug openings within 4 hours after engine shut-down without rotating the crankshaft so that cylinder bores, cylinder heads, and piston heads shall be evenly coated with undiluted preservative compound conforming to MIL-C-5545. If specified (see 6.2), corrosion-preventive used shall conform to MIL-C-6529, Type II. The cylinder preservative compound shall be sprayed at a nozzle temperature of 93.3° to 104.4°C. If aspirated with air, the air must be dry, and the fog temperature at the nozzle shall not be below 200°F. A relatively thin even coating of the cylinder preservative compound must be obtained on all the interior surfaces of the cylinder bore, cylinder heads, piston heads, and exposed portions of valves. The technique, equipment, and conditions incident to the spraying shall be subject to the approval of the Government inspector. CAUTION: If by accident the shaft is rotated, the cylinders shall be resprayed to insure unbroken coverage of the cylinder preservative compound on all surfaces. The aggregate volume of preservative compound from all spraying shall be not more than 20 percent of the volume of the cylinder with the piston at top center.

3.3.2.5 Closure of spark-plug openings. The spark plug openings of all engines shipped in dehydrated metal containers shall be closed by a suitable ventilatory plug.



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3.3.2.6 Ignition lead cleaning and protection. Spark plug lead terminals shall be cleaned using a lint-free cloth moistened with dry cleaning solvent conforming to P-D-680. The solvent shall not be permitted to run down inside the sleeves or spark plug leads. After cleaning and drying the terminal, clean protective caps conforming to AN 4060-1 or AN 4060-3 shall be installed.

3.3.2.7 Exhaust system. Each exhaust port or stack opening shall be closed with a suitable ventilatory plug or cover plate. Whenever engines are removed from dehydrated protection, the replacement of ventilated closures is mandatory unless the engine is to be operated within 24 hours of its removal from the dehydrated environment.

3.3.2.8 Fuel feed valves. Fuel feed valves or spinner injection type valves shall be preserved by draining excess fuel and forcing clean, filtered MIL-L-6082, Grade 1100 oil at 20 to 25 psi through the fuel feed valve or spinner injection nozzle until all residual fuel has been displaced and all surfaces normally in contact with fuel have been thoroughly coated with oil. The filter system used for assuring clean 1100 oil, shall effectively remove particles down to 5 microns in size. During the flushing operation the engine shall be in such a position that the displaced fuel and oil will run out the supercharger drain valve or carburetor air intake to avoid washing of the preservative coating from the cylinder bores.

3.3.2.9 Intake system. A solid cover shall be used temporarily to cover the carburetor air intake opening to protect the engine induction system against the entrance of moist air between the time the carburetor is removed and the engine is placed in a dehydrated metal container. The manifold opening shall be closed with a suitable ventilatory cover before the engine is placed in the dehydrated container.

3.3.2.10 Thrust bearings. Thrust bearings shall be preserved within 24 hours after the preservation run. For engines equipped with lubricator fittings or plugged slushing holes at the thrust bearing cover plates, runup compound shall be forced through these fittings or holes as appropriate. If lubricator fittings or slushing holes are not provided, the cover plate shall be loosened or removed and the thrust bearing shall be spray coated with the applicable runup compound. The plate shall then be reinstalled and secured.

3.3.2.11 Propeller shaft, propeller hub parts, and external bright metal surfaces. The exposed surfaces of the propeller shaft and other external bright metal surfaces shall be cleaned. Precautions shall be taken to remove fingerprints. These surfaces shall then be coated thoroughly with soft-film, corrosion-preventive compound conforming to MIL-C-16173, Grade 2. The propeller shaft shall then be wrapped with Grade A paper conforming to MIL-B-121 and secured with moisture-resistant tape conforming to PPP-T-60,



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or protected with an appropriate size plastic protector conforming to MS 27423. A propeller thread cap conforming to MS 27417, shall be placed over the end of the shaft. If a hoisting eye cap is required to place the engine in a shipping container, the installation of the shipping thread protector shall be delayed until after the engine is secure in the base of the container.

3.3.3 Final preparation of engines for shipment or storage.

3.3.3.1 Openings on engine. All openings on the engine not yet covered shall be closed using applicable ventilating type closures and methods.

3.3.3.2 Exterior cleaning of engines. Excess grease and oil and retained soils shall be removed from external surfaces. This cleaning shall be accomplished by solvent spray cleaning using dry cleaning solvent conforming to P-D-680.

3.3.3.3 Preservation tag. The proper preservation tag with correct preservation information filled in shall be attached (see Figure 2).

3.3.4 Recording of preservation. Completion of preservation shall be noted by correctly entering the information on the engine preservation record (see Figure 3).

3.3.5 Preservation marking. The following markings shall be applied to each container adjacent to or on the same side as the service receptacle as space permits.

The following, or any portion thereof, if applicable:

"THE CARBURETOR, WATER INJECTION EQUIPMENT, MANIFOLD PRESSURE REGULATOR, FUEL PUMP, AND PRIMER LINES ARE INCLUDED HEREIN."

The following additional information shall also be applied:

"THIS ENGINE IS PRESERVED BY THE USE OF DEHYDRATING AGENT."

"DO NOT BREAK THE SEALS ON THIS CONTAINER UNTIL READY FOR USE UNLESS INSPECTION OF THE INTERIOR HUMIDITY INDICATOR(S) SHOWS RENEWAL OF DEHYDRATING AGENT OR PRESERVATION TO BE NECESSARY."

"RELEASE AIR PRESSURE BEFORE LOOSENING BOLTS."

3.3.5.1 Preservation data. A space for application of preservation and represervation data, approximately 3 square feet with a contrasting background shall be provided. On Style 1 vertical containers, it shall be on the upper half of container and located relative to skid position and inspection ports, being visible and accessible under normal



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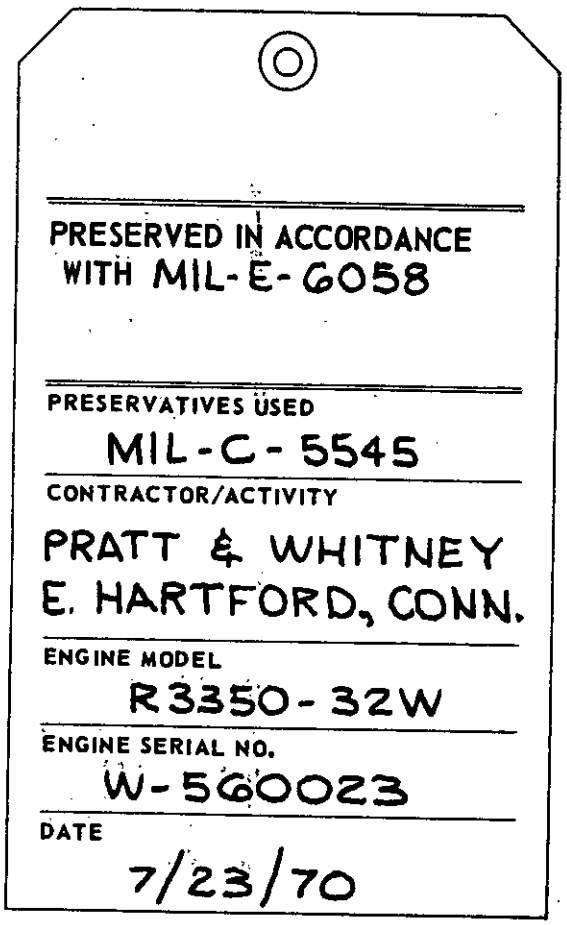
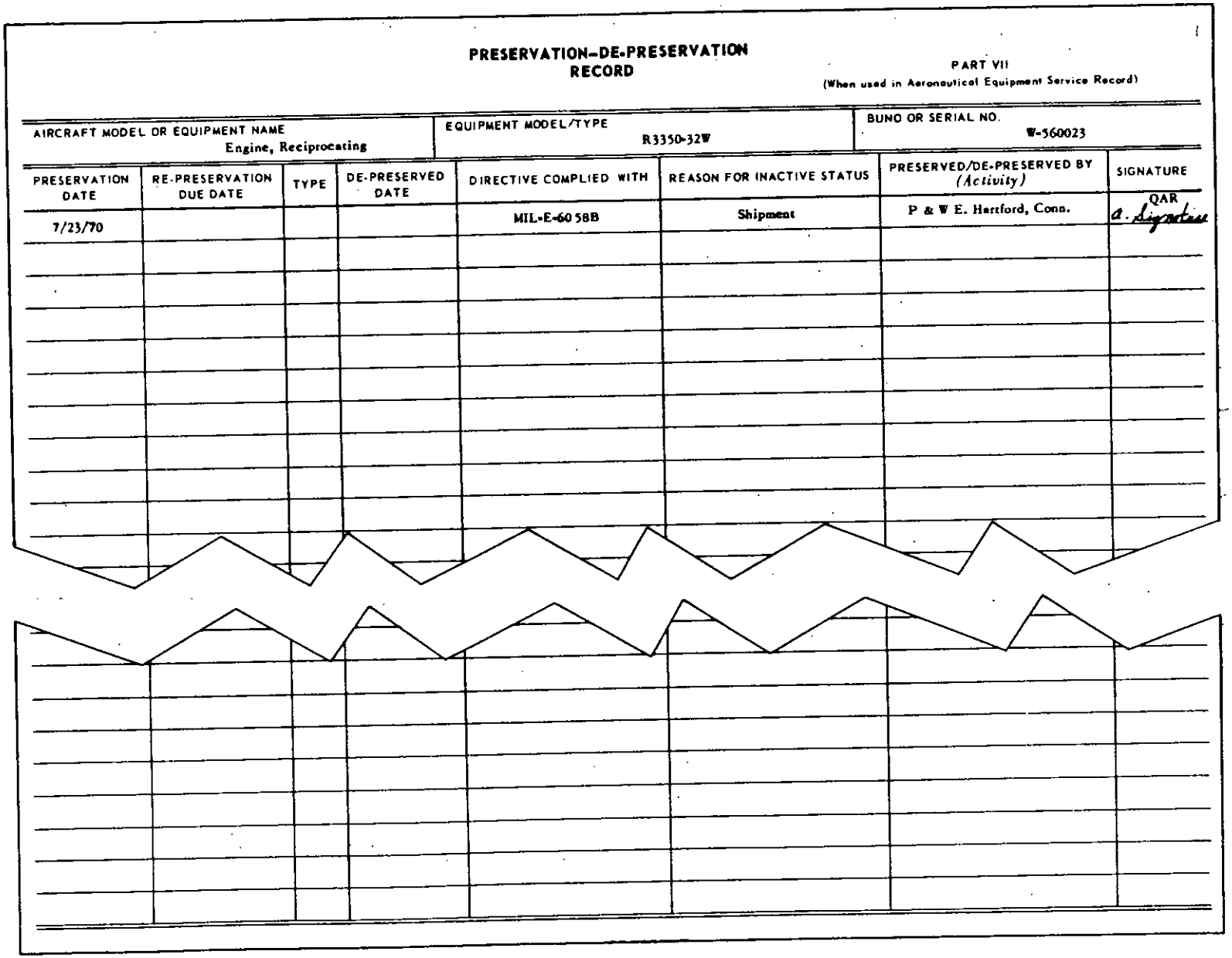



Figure 2. Example of properly filled out reciprocating engine preservation



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**PRESERVATION–DE-PRESERVATION RECORD**

PART VII
(When used in Aeronautical Equipment Service Record)

| AIRCRAFT MODEL OR EQUIPMENT NAME | EQUIPMENT MODEL/TYPE | BUNO OR SERIAL NO. |
|---|---|---|
| Engine, Reciprocating | R3350-32W | W-560023 |

| PRESERVATION DATE | RE-PRESERVATION DUE DATE | TYPE | DE-PRESERVED DATE | DIRECTIVE COMPLIED WITH | REASON FOR INACTIVE STATUS | PRESERVED/DE-PRESERVED BY (Activity) | SIGNATURE |
|---|---|---|---|---|---|---|---|
| 7/23/70 | | | | MIL-E-6058B | Shipment | P & W E. Hartford, Conn. | QAR a. Signature |

Figure 3. Example of correct preservation entry for reciprocating engines

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storage patterns. On Style II horizontal containers it shall be on the end of the container, adjacent to the inspection port. This information may be located adjacent to the same area reserved for necessary shipping data. The following data shall be indicated in at least 1/2-inch letters with space for fill-in of pertinent preservation information and inspection records:

PRESERVATION DATA

(a) Preservation type.

(b) Covering specification.

(c) Name of preserving activity.

(d) Date of preservation.

(e) Status of engine.

(f) Engine model.

(g) Engine serial number.

(h) A minimum of 6 spaces for inspection records labeled and titled as follows:

INSPECTION DATA

| Activity | Date |
|---|---|
| --- | --- |

3.4 Packing. Engines, preserved per 3.3, shall be packed in reusable metal containers conforming to MIL-C-5584 and in accordance with MIL-STD-794.

3.4.1 New containers. First article inspection requirements herein, where they pertain to packing, shall be governed by the requirements of MIL-C-5584 for the prototype container. When tested in accordance with Section 4 of MIL-C-5584, the container shall isolate the engine from g forces in excess of those specified as maximum allowable by the engine manufacturer. The relative humidity indicator required by MIL-C-5584 shall conform to MIL-I-26860, Type II. When the development of the container is sub-contracted by the engine manufacturer, and supervised by the engine manufacturer, the preliminary first article tests shall not be required as to the Government. When the development is sub-contracted by the engine manufacturer, it shall be sufficient for the requirement herein,



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for the engine manufacturer to submit the final test which will establish that the container conforms to M. quirements of this paragraph for the given engine. The panying the final test report shall be as specified on th Form 1423 made a part of the contract (see 6.2).

3.4.2 Pre-existing containers. When a container . ists which can be used as is, or with modification, for the giv the procuring activity shall so notify the engine manufacturer, a full details as to identification, availability, adaptation and qu cation of such container for use (see 6.2).

3.4.3 Installation in container.

(a) Follow the instructions accompanying the container for installing the engine in the container.

(b) The sling or lifting eye shall be removed from the engine.

(c) All spark plug leads or flexible lines shall be secured to a convenient surface using nylon cord conforming to MIL-C-5040 or pressure sensitive tape conforming to PPP-T-60.

(d) A warning placard (see Figure 4) shall be installed to call attention to the presence of ventilators in the engine openings.

(e) Accessory and Component Service Records and other applicable forms shall be placed in the log book binder and the binder shall be placed in the forms receptacle provided on the engine container.

(f) The appropriate amount of nonindicating desiccant, conforming to MIL-D-3464, Class 1, shall be placed in the desiccant basket provided in the shipping container. The amount of desiccant required shall be calculated using the following formula:

Units of desiccant = KV + XD

Legend for above formula:

K = 0.0007 when volume is given in cubic inches and 1.2 when given in cubic feet
V = volume within the barrier in cubic inches or cubic feet



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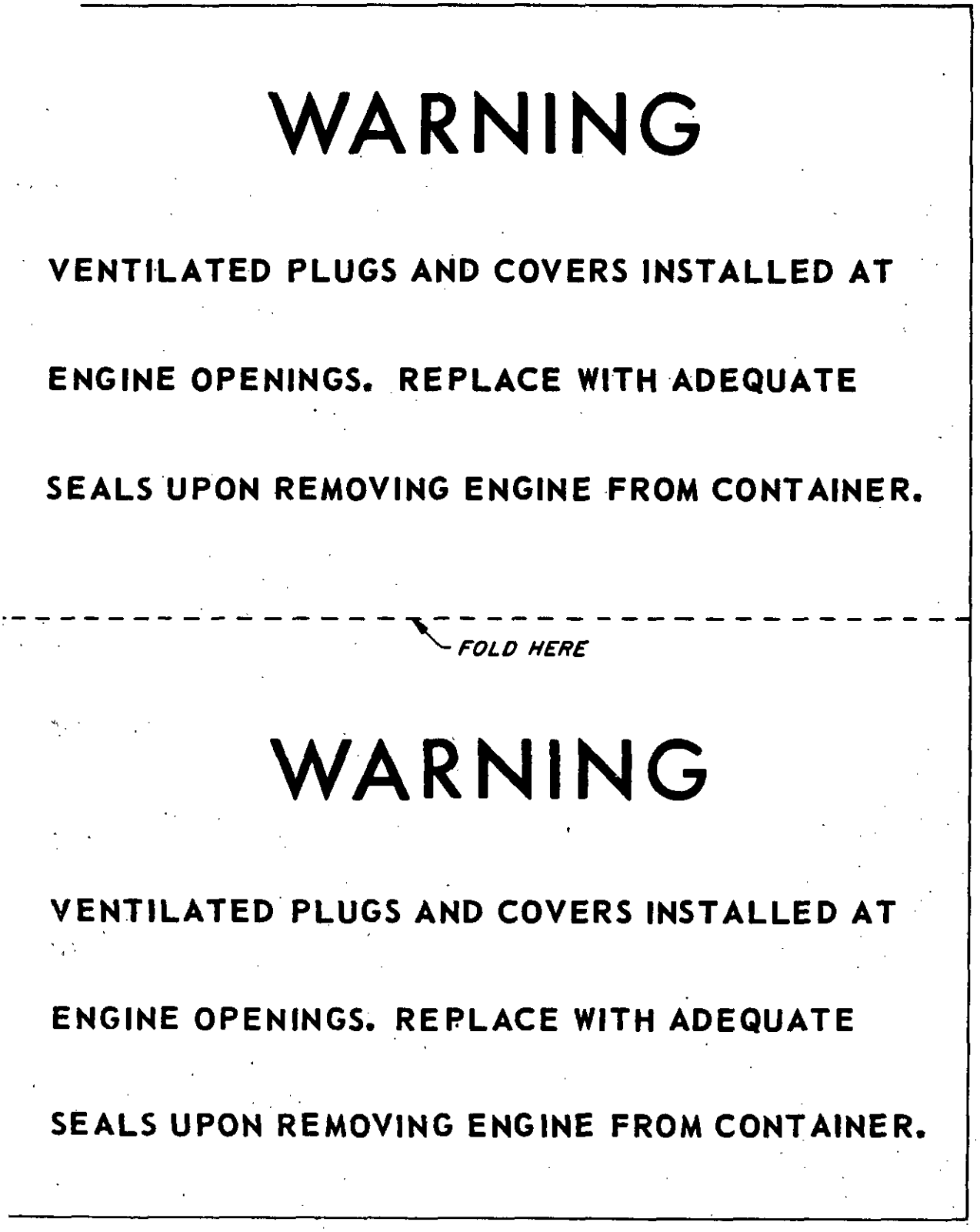



Figure 4. Warning placard, screened plug



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D = the pounds of dunnage other than metal within the barrier

The value of X varies with the type of dunnage or cushioning material used within the barrier as follows:

X = 8 for hair felt, cellulosic material (including wood) and other material not categorized below.
X = 6 for bound fibers (animal hair, synthetic fiber or vegetable bound with rubber).
X = 2 for glass fiber
X = 0.5 for synthetic foams and rubber

(g) Two tamperproof lead seals conforming to MS 51938 shall be attached to lid attaching bolts spaced 180 degrees apart. Installation of the seals shall be accomplished by substituting two drilled AN bolts for the regular lid attaching bolts.

3.5 Marking for shipment. In addition to the special requirements of the contract or order (see 6.2), shipments shall be marked in accordance with MIL-STD-129 and as follows:

(a) Destination.

(b) Shipping activity.

(c) Number of units.

(d) Invoice number.

(e) Weight.

(f) Cube.

(g) Dimensions.

(h) Bill of lading.

(i) Priority indicator.

(j) DMRI (date material required information).

(k) Command identification symbol.



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. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for he performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may se his own or any other facilities suitable for the performance of the nspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed ecessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)

(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 First article inspection samples. The first article ıspection samples shall conist of two complete packs that are representative of production items. The supplier shall conduct tests on the complete packaging systems to ascertain that the preservation, packing, and ırking of the unit (engine) conforms to this specification.

4.4 Quality conformance tests. Sample packs, containing ıgines packed as for shipment, shall be selected and inspected in acordance with the following examinations and tests performed in the order sted.

(a) Examination of Individual packs (4.4.2)

(b) Leak test (4.5)

(c) Examination of Quality Conformance Lots (4.4.3)

4.4.1 Lot. All packs produced during one work day (8 hours) ıall be considered a lot.

4.4.2 Visual examination of individual packs. A thorough sual examination of each pack shall be conducted with emphasis on the llowing points as listed in Table I.



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TABLE I

LIST OF DEFECTS FOR VISUAL EXAMINATION OF INDIVIDUAL CONTAINERS *

| Examine | Defect |
|---|---|
| Container (exterior) | Container does not conform to MIL-C-5584. |
| Markings | Any preservation marking missing, incomplete, incorrect, improperly applied or obliterated. |
| Closure | Any fastener, cover or cap, lead tamper proof seal missing or improperly applied. Flange bolts not preservative coated. Flange gasket improperly seated or damaged. |
| Records receptacle | Records missing, incomplete, incorrectly filled out. Receptacle not properly closed. |

* For Army see TB55-8100-200-25.

4.4.2.1 Re-examination. Containers held more than one week before shipment to first receiving activity shall be re-examined according to 4.4.2 every seven days.

4.4.3 Visual examination for quality conformance lots. The visual examinations shall consist of all the examinations listed in Table II.

4.4.4 Sampling for tests. Sampling for Leak test shall be at Inspection Level S-4 of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 0 percent defective. The unit of sampling shall be one pack.

4.5 Leak test. Containers shall be tested for leaks by pressurizing to 5 psi, and applying a soap solution to all joints and weld seams. If necessary, the pressure relief valve shall be removed and the aperture blocked off. At the conclusion of the test the internal pressure shall be vented off to equalize internal and external pressure.

4.6 Cause for rejection. At the conclusion of the above test, the condition of the containers and engine shall be determined in accordance with Table II.



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TABLE II

CLASSIFICATION AND LIST OF DEFECTS FOR VISUAL EXAMINATION OF QUALITY CONFORMANCE LOTS

| Examine | Requirement Paragraph | Defect | Major | Minor |
|---|---|---|---|---|
| Container (exterior) | | | | |
| (a) Markings | 3.3.5 | Any marking missing, incomplete, incorrect, improperly applied or obliterated. | X 1/ | |
| (b) Closure | 3.4.3 (g) | 1. Any fastener, cover or cap, lead tamper proof seal missing or improperly applied. | X | |
| | | 2. Flange bolts not preservative coated. | | X |
| | | 3. Flange gasket improperly seated or damaged. | X | |
| (c) Records receptacle | 3.4.3 (e) | 1. Records missing, incomplete, incorrectly filled out. | X 1/ | |
| | | 2. Receptacle not properly closed. | | X |
| Container (interior) | | | | |
| (d) Installation | 3.4.3 | Fasteners or attaching hardware incorrect, missing, improperly attached, incorrectly aligned. Locking devices missing where required. | X | |
| (e) Warning placard | 3.4.3 (d) | Ventilated plug warning placard not visible or missing. | | X 2/ |
| (f) Flexible lines and spark plug leads | 3.4.3 (c) | Not secured. | X | |
| (g) Desiccant | 3.4.3 (f) | Missing. Incomplete application. Incorrect type. Not properly distributed. | X | |
| (h) Cleanness | 3.3.3.2 | Not cleaned. | | X |
| (i) Sling or lifting eye | 3.4.3 (b) | Not removed. | | X |
| (j) Preservative tag | 3.3.3.3 | 1. Missing | X | |
| | | 2. Not visible. Not securely attached. | | X |
| | | 3. Incorrectly filled out. | | X 3/ |
| (k) Exterior closures and caps | 3.3.3.1 | Any cap, closure or cover missing, incorrect, improperly applied. | | X 4/ |
| (l) Accessories and spark plugs | 3.3.2.1 | Any accessory or spark plug not removed. | X | |
| (m) Accessory drives | 3.3.2.2 | 1. Not preservative coated. | X | |
| | | 2. Covers missing. | X | |
| | | 3. Not safetied. | | X |



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TABLE II (Continued)

| Examine | Requirement Paragraph | Defect | Major | Minor |
|---|---|---|---|---|
| (n) Ignition leads | 3.3.2.6 | Not cleaned and capped. | | X |
| (o) Thrust bearings | 3.3.2.10 | 1. Not properly preserved. | X | |
| | | 2. Cover plates missing. | X | |
| | | 3. Cover plates improperly secured. | | X |
| (p) Cylinder bores | 3.3.2.4 | Not preserved, uneven preservative coating, excessive preservative deposit when inspected with borescope. | X | |

1/ If easily and correctly rectifiable the defect shall be classified as minor.
2/ This defect shall be classified as major if in conjunction with (a).
3/ This defect shall be classified as major if in conjunction with (a), (c), or (e).
4/ This defect shall be classified as major if in conjunction with any other minor defect.



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## 5. PREPARATION FOR DELIVERY

5.1 Not applicable.

## 6. NOTES

6.1 Intended use. The preservation and packing system for aircraft reciprocating engines detailed by this specification, is intended for use by manufacturers and activities engaged in the preservation and shipment of newly manufactured or completely overhauled engines. Engines preserved according to this specification normally require only inspection for maintenance of container pressure and safe relative humidity conditions once every 90 days during indefinite storage.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Any additional marking required (see 3.5).

(c) Any deviations from requirements of this specification permitted by the procuring activity.

(d) Any additional requirements.

(e) Method of cleaning, if other than C-1 of MIL-P-116.

(f) DD Form 1423 requirements for container drawings (see 3.4.1).

(g) Special instructions on existing containers (see 3.4.2).

(h) Any special marking requirements (see 3.5).

(i) If MIL-C-6529, Type II, corrosion-preventive shall be used (see 3.3.1.2 and 3.3.2.4).

6.3 Data. For the information of Contractors and Contracting Officers, any of the data specified in (a) applicable documents listed in Section 2 of this specification or (b) referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List) which shall be attached to and made a part of the contract or order.



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Custodians:
Army - AV
Navy - AS
Air Force - 82

Preparing activity:
Navy - AS
(Project No. PACK-0244)



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